

TATA CONSULTANCY SERVICES(TCS iON)

Digital Learning Program



INDUSTRY PROJECT REPORT

Phishing Attack Awareness and Defense

For Campus Facilities

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ABSTRACT

Phishing is one of the most common cyber threats used by attackers to steal sensitive information such as usernames, passwords, and banking details. These attacks are carried out using fake emails, messages, and websites that appear legitimate.

This project focuses on understanding phishing attacks, identifying red flags, and demonstrating how such attacks work using a simulated login page. A simple web-based phishing demo was developed using HTML and JavaScript to show how user credentials can be captured.

The results show that phishing attacks can be easily performed if users are not aware, highlighting the importance of cybersecurity awareness and preventive measures.

CHAPTER 1: INTRODUCTION

1.1 Background

With the increasing use of the internet for banking, shopping, and communication, cyber threats like phishing have become very common. Attackers use fake emails and websites to trick users into sharing confidential information.

1.2 Problem Statement

Many users are unaware of phishing attacks and fall victim to fake messages and websites. This leads to data theft, financial loss, and security issues.

1.3 Objective

- * To understand phishing attacks
- * To identify phishing red flags
- * To demonstrate phishing using a fake login page
- * To suggest prevention techniques

1.4 Scope of the Project

This project focuses on phishing awareness and simulation using a simple web-based login page. It does not involve real hacking or real data collection.

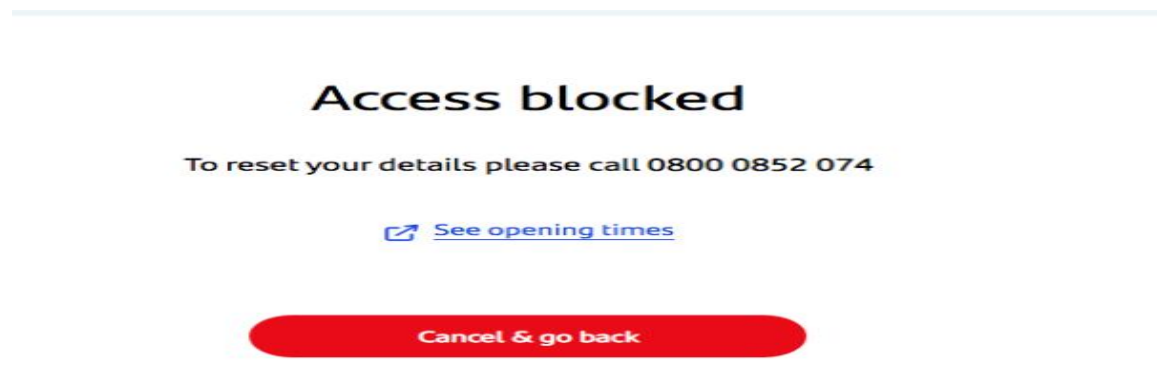
CHAPTER 2: MATERIALS AND METHODS

2.1 Tools and Technologies

- * HTML (for login page)
- * CSS (for styling)
- * JavaScript (for capturing input)
- * Web Browser (Chrome / Edge)

2.2 Phishing Example

A phishing example was used showing a “Bank Account Blocked” message, which creates urgency and fear in users.



2.3 Phishing Red Flags

- * Unknown sender
- * Urgent message
- * Suspicious links
- * Fake website design
- * Request for sensitive data

2.4 Suspicious Link Example

www.secure-bank-login-alert.com

This link appears real but is actually fake.

2.5 Phishing Simulation

A fake login page was created where users enter:

- * Username

- * Password

The entered data is captured and displayed using JavaScript.

CHAPTER 3: TESTING AND VALIDATION

3.1 Introduction to Testing

Testing ensures that the phishing simulation works correctly.

3.2 Functional Testing

- * Login page loads correctly ✓
- * Input fields work properly ✓
- * Data is captured successfully ✓

3.3 Output Testing

When user enters details:

- * Username and password are displayed in popup

3.4 User Testing

Users tested the system and understood how phishing works.

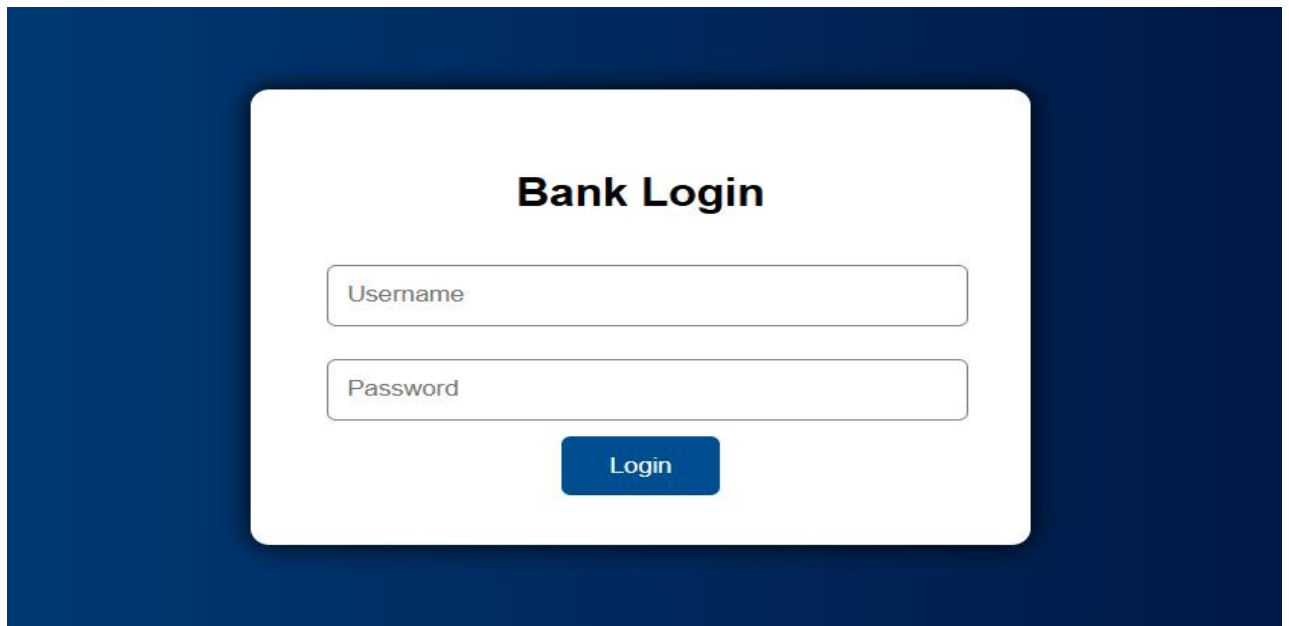
3.5 Result

The system successfully demonstrates phishing behaviour.

CHAPTER 4: RESULTS

4.1 System Interface

Figure 4.1: Login Page

A screenshot of a web page with a dark blue background. In the center is a white rounded rectangle containing the text "Bank Login" in bold. Below this text are two input fields: the first is labeled "Username" and the second is labeled "Password". Below the input fields is a blue button with the text "Login" in white.

Bank Login

Username

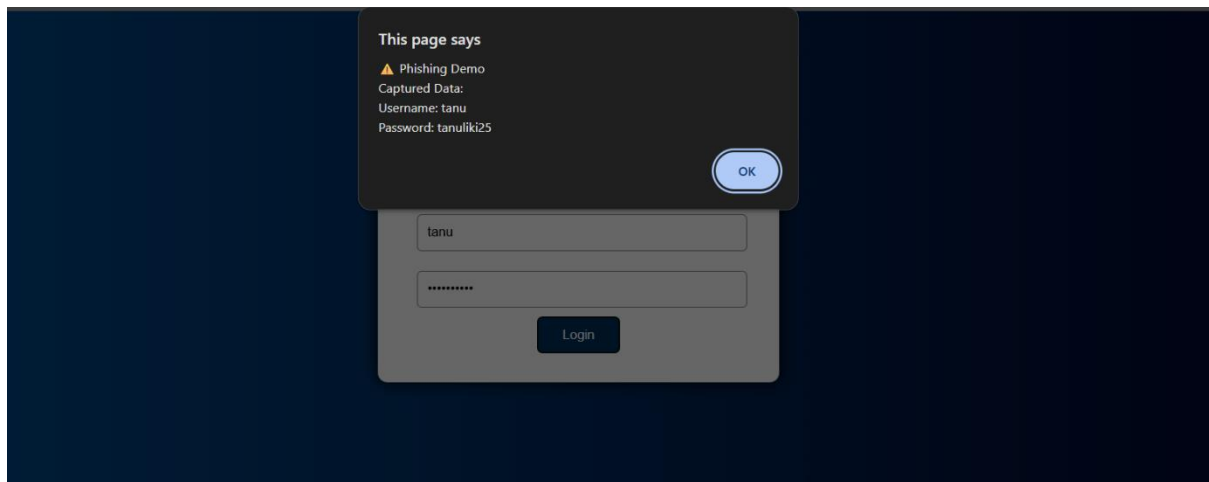
Password

Login

Description:

This shows the fake bank login page used for phishing simulation.

Figure 4.2: Captured Data Output



Description:

This shows how user credentials are captured.

4.2 Results Analysis

The project successfully demonstrates how phishing attacks trick users into entering sensitive information. It shows that users can be easily misled by fake login pages and urgent messages.

CHAPTER 5: CONCLUSION

Phishing is a serious cybersecurity threat that can lead to data theft and financial loss. This project demonstrates how phishing attacks work and highlights the importance of user awareness.

By identifying red flags and following safe practices, users can protect themselves from phishing attacks.

FUTURE WORK

- * Add real-time phishing detection
- * Use AI to detect phishing emails
- * Create awareness campaigns
- * Improve UI design